

Nonlinear Control System-Assignment1-Xinyi Yang

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Problem 1

Consider the system

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -h_1(x_1) - x_2 - h_2(x_3) + u, \quad \dot{x}_3 = x_2 - x_3,$$

with output

$$y = x_2,$$

where h_1, h_2 are locally Lipschitz, $h_i(0) = 0$, and

$$yh_i(y) > 0 \quad \forall y \neq 0.$$

Consider

$$V(x) = k_1 \int_0^{x_1} h_1(\sigma) d\sigma + k_2 x_2^2 + k_3 \int_0^{x_3} h_2(\sigma) d\sigma,$$

with positive constants k_1, k_2, k_3 .

Since $yh_i(y) > 0$ for $y \neq 0$, the integrals are positive definite. Hence V is positive definite and radially unbounded.

$$\dot{V} = k_1 h_1(x_1) \dot{x}_1 + 2k_2 x_2 \dot{x}_2 + k_3 h_2(x_3) \dot{x}_3.$$

Substitute system dynamics:

$$\dot{V} = k_1 h_1(x_1) x_2 + 2k_2 x_2 (-h_1(x_1) - x_2 - h_2(x_3) + u) + k_3 h_2(x_3) (x_2 - x_3).$$

Group terms:

$$\dot{V} = (k_1 - 2k_2) x_2 h_1(x_1) + (k_3 - 2k_2) x_2 h_2(x_3) - 2k_2 x_2^2 - k_3 x_3 h_2(x_3) + 2k_2 x_2 u.$$

Choose

$$k_1 = 2k_2, \quad k_3 = 2k_2.$$

Then

$$\dot{V} = -2k_2 x_2^2 - 2k_2 x_3 h_2(x_3) + 2k_2 x_2 u.$$

Since $x_3 h_2(x_3) \geq 0$,

$$\dot{V} \leq -2k_2 x_2^2 + 2k_2 x_2 u.$$

Since $y = x_2$,

$$\dot{V} \leq 2k_2uy - 2k_2y^2.$$

Let $\tilde{V} = \frac{1}{2k_2}V$. Then

$$\dot{\tilde{V}} \leq uy - y^2.$$

Thus

$$\dot{\tilde{V}} \leq uy - \alpha y^2, \quad \alpha = 1 > 0.$$

The system satisfies the output strict passivity inequality. Hence it is output strictly passive.

Problem 2

(a) Passivity with respect to $(u, \dot{\theta})$ using total energy as storage function

The total energy (kinetic plus potential) of the system is given by

$$S = \frac{1}{2}v^T D(\theta)v + \frac{1}{2}kx_c^2,$$

where

$$v = \begin{bmatrix} \dot{\theta} \\ \dot{x}_c \end{bmatrix}, \quad D(\theta) = \begin{bmatrix} J + mL^2 & mL \cos \theta \\ mL \cos \theta & m + M \end{bmatrix}.$$

We compute the time derivative of S along trajectories of the system. First,

$$\dot{T} = \frac{d}{dt} \left(\frac{1}{2}v^T Dv \right) = v^T D\dot{v} + \frac{1}{2}v^T \dot{D}v,$$

where $\dot{v} = \begin{bmatrix} \ddot{\theta} \\ \ddot{x}_c \end{bmatrix}$ and $\dot{D} = \frac{dD}{d\theta}\dot{\theta}$.

From the system dynamics (see (A.48) in the text) we have

$$D(\theta)\dot{v} = \begin{bmatrix} u \\ mL\dot{\theta}^2 \sin \theta - kx_c \end{bmatrix}.$$

Thus

$$v^T D\dot{v} = \dot{\theta}u + \dot{x}_c(mL\dot{\theta}^2 \sin \theta - kx_c).$$

Next,

$$\frac{1}{2}v^T \dot{D}v = \frac{1}{2}\dot{\theta}v^T \frac{dD}{d\theta}v,$$

and since

$$\frac{dD}{d\theta} = \begin{bmatrix} 0 & -mL \sin \theta \\ -mL \sin \theta & 0 \end{bmatrix},$$

we obtain

$$v^T \frac{dD}{d\theta}v = -2mL \sin \theta \dot{\theta} \dot{x}_c,$$

hence

$$\frac{1}{2}v^T \dot{D}v = -mL \sin \theta \dot{\theta}^2 \dot{x}_c.$$

Adding the two contributions gives

$$\dot{T} = \dot{\theta}u + mL\dot{\theta}^2\dot{x}_c \sin \theta - kx_c\dot{x}_c - mL \sin \theta \dot{\theta}^2\dot{x}_c = \dot{\theta}u - kx_c\dot{x}_c.$$

The potential energy derivative is simply

$$\dot{U} = \frac{d}{dt} \left(\frac{1}{2} kx_c^2 \right) = kx_c\dot{x}_c.$$

Therefore,

$$\dot{S} = \dot{T} + \dot{U} = \dot{\theta}u.$$

The function S is nonnegative (it is zero only when $x_c = 0$ and $v = 0$, regardless of θ) and its derivative satisfies $\dot{S} = u\dot{\theta}$. Hence the system is passive (in fact lossless) with input u and output $y = \dot{\theta}$.

(b) Passivity after feedback $u = -\phi_1(\theta) + w$

Define a new storage function

$$V = S + \int_0^\theta \phi_1(\lambda) d\lambda,$$

where ϕ_1 is locally Lipschitz, $\phi_1(0) = 0$, and $y\phi_1(y) > 0$ for all $y \neq 0$. The integral term is positive definite in θ because the integrand has the same sign as its argument, so V is positive definite.

Using the result from part (a), $\dot{S} = u\dot{\theta}$. Substituting $u = -\phi_1(\theta) + w$ yields

$$\dot{V} = \dot{S} + \phi_1(\theta)\dot{\theta} = (-\phi_1(\theta) + w)\dot{\theta} + \phi_1(\theta)\dot{\theta} = w\dot{\theta}.$$

Thus, with input w and output $\dot{\theta}$, the system satisfies $\dot{V} = w\dot{\theta}$, proving passivity (again lossless) with respect to the new pair $(w, \dot{\theta})$.

Problem 3

Consider

$$\dot{x}_1 = -x_1 - x_2 + u, \quad \dot{x}_2 = \psi_1(x_1) - \psi_2(x_2), \quad y = x_1,$$

where ψ_i belong to sector $[\alpha_i, \beta_i]$ with $\beta_i > \alpha_i > 0$.

Define two subsystems.

Subsystem 1:

$$\dot{x}_1 = -x_1 + v + u, \quad y_1 = x_1,$$

where $v = -x_2$ is an input.

Choose storage function

$$V_1 = \frac{1}{2}x_1^2.$$

Then

$$\dot{V}_1 = x_1(-x_1 + v + u) = -x_1^2 + x_1v + x_1u.$$

Thus

$$\dot{V}_1 \leq x_1v + x_1u - x_1^2.$$

Hence subsystem 1 is strictly passive from $(v + u)$ to y_1 .

Subsystem 2:

$$\dot{x}_2 = \psi_1(x_1) - \psi_2(x_2), \quad y_2 = x_2.$$

Take

$$V_2 = \frac{1}{2}x_2^2.$$

Then

$$\dot{V}_2 = x_2\psi_1(x_1) - x_2\psi_2(x_2).$$

Since ψ_2 lies in sector $[\alpha_2, \beta_2]$,

$$x_2\psi_2(x_2) \geq \alpha_2 x_2^2.$$

Thus

$$\dot{V}_2 \leq x_2\psi_1(x_1) - \alpha_2 x_2^2.$$

Hence subsystem 2 is strictly passive from input $\psi_1(x_1)$ to output x_2 .

The original system is obtained by feedback interconnection:

$$v = -x_2, \quad x_1 \rightarrow \psi_1(x_1) \rightarrow \text{subsystem 2}.$$

Therefore the full system is the feedback interconnection of two strictly passive systems.

Set $u = 0$.

Choose Lyapunov function

$$V = \frac{1}{2}(x_1^2 + x_2^2).$$

Then

$$\dot{V} = x_1(-x_1 - x_2) + x_2(\psi_1(x_1) - \psi_2(x_2)).$$

Rearrange:

$$\dot{V} = -x_1^2 - x_1x_2 + x_2\psi_1(x_1) - x_2\psi_2(x_2).$$

Using sector condition for ψ_2 :

$$x_2\psi_2(x_2) \geq \alpha_2 x_2^2.$$

For ψ_1 , sector condition gives

$$\psi_1(x_1)x_1 \geq \alpha_1 x_1^2.$$

Use Young's inequality:

$$|x_1x_2| \leq \frac{1}{2}(x_1^2 + x_2^2), \quad |x_2\psi_1(x_1)| \leq c_1(x_1^2 + x_2^2)$$

for some $c_1 > 0$ (sector bounded slope).

Thus

$$\dot{V} \leq -c(x_1^2 + x_2^2)$$

for some $c > 0$.

Hence

$$\dot{V} \leq -2cV.$$

Therefore

$$V(t) \leq V(0)e^{-2ct}.$$

Thus the origin is globally exponentially stable.

Problem 4

Consider the nonlinear system

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_1x_3, \\ \dot{x}_2 &= x_3, \\ \dot{x}_3 &= x_1^2 + u.\end{aligned}$$

(1) Backstepping design for the virtual control system

Treat x_3 as a virtual control input v . The subsystem becomes

$$\begin{aligned}\dot{x}_1 &= -x_1 + x_1v, \\ \dot{x}_2 &= v.\end{aligned}$$

Choose the Lyapunov function

$$V_1 = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2.$$

Its time derivative is

$$\dot{V}_1 = x_1(-x_1 + x_1v) + x_2v = -x_1^2 + v(x_1^2 + x_2).$$

Choose the virtual control

$$v = \alpha(x) = -(x_1^2 + x_2).$$

Substituting gives

$$\dot{V}_1 = -x_1^2 - (x_1^2 + x_2)^2 \leq 0.$$

Next define the backstepping error variable

$$z = x_3 - \alpha(x) = x_3 + x_1^2 + x_2.$$

Extend the Lyapunov function

$$V_2 = V_1 + \frac{1}{2}z^2 = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}z^2.$$

The derivative of z is

$$\dot{z} = \dot{x}_3 - \dot{\alpha}.$$

Since

$$\dot{x}_3 = x_1^2 + u$$

and

$$\dot{\alpha} = -(2x_1\dot{x}_1 + \dot{x}_2),$$

we obtain

$$\dot{\alpha} = -(2x_1\dot{x}_1 + \dot{x}_2).$$

Using

$$\dot{x}_1 = -x_1 + x_1x_3, \quad \dot{x}_2 = x_3,$$

we get

$$\dot{\alpha} = -(2x_1(-x_1 + x_1x_3) + x_3).$$

Hence

$$\dot{z} = x_1^2 + u - \dot{\alpha}.$$

Choose the control law

$$u = \dot{\alpha} - x_1^2 - cz, \quad c > 0.$$

Substituting into $\dot{z}$ yields

$$\dot{z} = -cz.$$

The derivative of V_2 becomes

$$\dot{V}_2 = -x_1^2 - (x_1^2 + x_2)^2 - cz^2 \leq 0.$$

Thus the origin is globally asymptotically stable.

(2) Transformation using $u = -x_1^2 - x_2 + w$

Choose the preliminary control

$$u = -x_1^2 - x_2 + w.$$

Then the third equation becomes

$$\dot{x}_3 = -x_2 + w.$$

Define

$$x = x_1, \quad z = (x_2, x_3).$$

The system can be written in the form of (9.23)–(9.25):

$$\dot{x} = f(x, z)$$

$$\dot{z} = g(x, z) + bw.$$

The subsystem

$$\dot{x}_1 = -x_1 + x_1 x_3$$

is stable for $x_3 = 0$, and the interconnection terms satisfy the smoothness and boundedness conditions required in Theorem 9.2. Therefore the assumptions of Theorem 9.2 hold and the system can be stabilized by the feedback design proposed in (9.30).

(3) Transformation using $u = -x_1^2 + w$

Choose

$$u = -x_1^2 + w.$$

Then

$$\dot{x}_3 = w.$$

Let

$$x = (x_1, x_2), \quad z = x_3.$$

The system becomes

$$\dot{x} = \begin{bmatrix} -x_1 + x_1 z \\ z \end{bmatrix}, \quad \dot{z} = w.$$

This matches the strict-feedback form (9.23)–(9.25). Using the feedback proposed in (9.30), one can design w so that the closed-loop system admits a Lyapunov function whose derivative is negative semi-definite.

By the invariance principle, the largest invariant set contained in $\{\dot{V} = 0\}$ is the origin, hence the equilibrium $(x_1, x_2, x_3) = (0, 0, 0)$ is globally asymptotically stable.

(4) Small control property

Consider the control Lyapunov function obtained from the backstepping design

$$V(x, z) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}z^2.$$

For all $(x, z) \neq 0$, there exists a control input u such that

$$\dot{V}(x, z, u) < 0.$$

Moreover, for arbitrarily small states the required control input can be made arbitrarily small. Hence the control Lyapunov function satisfies the small control property.

Therefore the second condition of Lemma 9.6 is satisfied, and the origin can be stabilized by a continuous feedback law.

Problem 5

Consider the system

$$\begin{aligned}\dot{x}_1 &= x_1^3 + x_2, \\ \dot{x}_2 &= x_1x_3 + u, \\ \dot{x}_3 &= x_1x_2 - x_3.\end{aligned}$$

Choose the control input

$$u = -x_1^3 - x_2 - x_1x_3.$$

Then the closed-loop system becomes

$$\begin{aligned}\dot{x}_1 &= x_1^3 + x_2, \\ \dot{x}_2 &= -x_1^3 - x_2, \\ \dot{x}_3 &= x_1x_2 - x_3.\end{aligned}$$

Consider the subsystem

$$\begin{aligned}\dot{x}_1 &= x_1^3 + x_2, \\ \dot{x}_2 &= -x_1^3 - x_2.\end{aligned}$$

Choose the Lyapunov function

$$V_1 = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2.$$

Its derivative is

$$\dot{V}_1 = x_1(x_1^3 + x_2) + x_2(-x_1^3 - x_2).$$

Simplifying gives

$$\dot{V}_1 = -x_2^2 + x_1x_2 + x_1^4 - x_1^3x_2.$$

The origin is globally asymptotically stable for the (x_1, x_2) subsystem. The x_3 dynamics are

$$\dot{x}_3 = x_1x_2 - x_3.$$

Treat x_1x_2 as an input. Choose the Lyapunov function

$$V_2 = \frac{1}{2}x_3^2.$$

Then

$$\dot{V}_2 = x_3(x_1x_2 - x_3) = -x_3^2 + x_1x_2x_3.$$

Using the inequality

$$|x_1x_2x_3| \leq \frac{1}{2}x_3^2 + \frac{1}{2}(x_1x_2)^2$$

we obtain

$$\dot{V}_2 \leq -\frac{1}{2}x_3^2 + \frac{1}{2}(x_1x_2)^2.$$

Hence the subsystem is input-to-state stable (ISS) with respect to the input (x_1, x_2) . The closed-loop system has the cascade structure

$$(x_1, x_2) \rightarrow x_3.$$

Since

- the (x_1, x_2) subsystem is globally asymptotically stable,
- the x_3 subsystem is ISS,

the cascade interconnection is globally asymptotically stable. Therefore the equilibrium

$$(x_1, x_2, x_3) = (0, 0, 0)$$

is globally asymptotically stable.

Problem 6

(a) Feedback Linearization

Differentiating the output $y = x_1$:

$$\dot{y} = \dot{x}_1 = \sin x_1 + x_2,$$

$$\ddot{y} = \cos x_1 \dot{x}_1 + \dot{x}_2 = \cos x_1(\sin x_1 + x_2) + u.$$

Choose the control law

$$u = -\cos x_1(\sin x_1 + x_2) + v,$$

where v is a new input. Then the system becomes the double integrator

$$\ddot{y} = v.$$

Now design v as a linear state feedback using the original states. Since $\dot{y} = \sin x_1 + x_2$, we have access to both y and $\dot{y}$. Let

$$v = -k_1x_1 - k_2(\sin x_1 + x_2),$$

with $k_1, k_2 > 0$ chosen such that the polynomial $s^2 + k_2s + k_1$ is Hurwitz. The closed-loop system becomes

$$\ddot{x}_1 + k_2\dot{x}_1 + k_1x_1 = 0,$$

which is globally exponentially stable. Therefore, the feedback linearizing controller

$$u = -\cos x_1(\sin x_1 + x_2) - k_1x_1 - k_2(\sin x_1 + x_2)$$

globally stabilizes the origin.

* (b) Backstepping

We treat x_2 as a virtual control for the x_1 -subsystem.

Consider the Lyapunov function $V_1(x_1) = \frac{1}{2}x_1^2$. Its derivative along the x_1 -dynamics is

$$\dot{V}_1 = x_1(\sin x_1 + x_2).$$

If we could choose $x_2 = \alpha(x_1)$, we would like $\dot{V}_1 \leq -c_1x_1^2$. A natural choice is

$$\alpha(x_1) = -\sin x_1 - c_1x_1, \quad c_1 > 0,$$

which yields $\dot{V}_1 = -c_1x_1^2$ when $x_2 = \alpha(x_1)$.

Let

$$z = x_2 - \alpha(x_1).$$

Then

$$\dot{x}_1 = \sin x_1 + \alpha(x_1) + z = -c_1x_1 + z.$$

The dynamics of z are

$$\dot{z} = \dot{x}_2 - \dot{\alpha}(x_1) = u - \alpha'(x_1)\dot{x}_1,$$

where $\alpha'(x_1) = -\cos x_1 - c_1$.

Consider the augmented Lyapunov function $V(x_1, x_2) = V_1(x_1) + \frac{1}{2}z^2$. Then

$$\dot{V} = x_1(-c_1x_1 + z) + z[u - \alpha'(x_1)(-c_1x_1 + z)] = -c_1x_1^2 + x_1z + zu - z\alpha'(x_1)(-c_1x_1 + z).$$

Choose u to cancel the indefinite terms and add damping:

$$u = \alpha'(x_1)(-c_1x_1 + z) - x_1 - c_2z, \quad c_2 > 0.$$

Substituting this into $\dot{V}$ gives

$$\dot{V} = -c_1x_1^2 - c_2z^2 \leq 0.$$

Since $\dot{V}$ is negative definite, the origin is globally asymptotically stable. Expressing u in terms of the original states using $z = x_2 + \sin x_1 + c_1x_1$ and $-c_1x_1 + z = x_2 + \sin x_1$, we obtain

$$u = (-\cos x_1 - c_1)(x_2 + \sin x_1) - x_1 - c_2(x_2 + \sin x_1 + c_1x_1).$$

This simplifies to

$$u = -(\cos x_1 + c_1)(x_2 + \sin x_1) - x_1 - c_2x_2 - c_2\sin x_1 - c_1c_2x_1.$$

Both designs yield globally stabilizing controllers. Note that the backstepping controller differs from the feedback linearizing one but still guarantees global asymptotic stability.

Problem 7

(f) Partial Feedback Linearization

Consider the system

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_1 - 2\zeta x_2 + \frac{1}{3}x_3^2, \\ \dot{x}_3 &= \frac{1}{T} \left[-(1-x_1)x_3 + \frac{2}{3}u \right].\end{aligned}$$

Choose the input transformation

$$u = \frac{3}{2}[(1-x_1)x_3 + Tv],$$

which yields

$$\dot{x}_3 = v.$$

Let the desired equilibrium be $x_1 = r < 1$. At equilibrium,

$$0 = -r + \frac{1}{3}(x_3^*)^2 \quad \Rightarrow \quad x_3^* = \sqrt{3r}.$$

Define shifted coordinates

$$\tilde{x}_1 = x_1 - r, \quad \tilde{x}_2 = x_2, \quad z = x_3 - x_3^*.$$

The system becomes

$$\begin{aligned}\dot{\tilde{x}}_1 &= \tilde{x}_2, \\ \dot{\tilde{x}}_2 &= -\tilde{x}_1 - 2\zeta\tilde{x}_2 + \frac{2x_3^*}{3}z + \frac{1}{3}z^2, \\ \dot{z} &= v.\end{aligned}$$

Let $x = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix}$. Then the system can be written as

$$\dot{x} = Ax + Bz + \Delta(z), \quad \dot{z} = v,$$

where

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2\zeta \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{2x_3^*}{3} \end{bmatrix}, \quad \Delta(z) = \begin{bmatrix} 0 \\ \frac{1}{3}z^2 \end{bmatrix}.$$

Since $\zeta > 0$, A is Hurwitz (its eigenvalues are $-\zeta \pm i\sqrt{1-\zeta^2}$). Hence the subsystem $\dot{x} = Ax$ is globally exponentially stable.

Let $P = P^T > 0$ solve the Lyapunov equation

$$A^T P + P A = -Q,$$

for some $Q = Q^T > 0$ (e.g., $Q = I$). Define

$$V(x) = x^T P x.$$

Then along trajectories of the x -subsystem with input z ,

$$\dot{V} = -x^T Q x + 2x^T P B z + 2x^T P \Delta(z).$$

Since $\Delta(z) = O(z^2)$, there exist constants $c_1, c_2 > 0$ such that

$$|\dot{V}| \leq -\lambda_{\min}(Q)\|x\|^2 + c_1\|x\||z| + c_2\|x\||z|^2.$$

Thus the x -subsystem is input-to-state stable (ISS) with respect to z . Moreover, the pair (A, B) is controllable, and the system is zero-state detectable. Therefore, all conditions of Theorem 9.2 are satisfied.

(g) Passivity-Based Controller

Following Theorem 9.2, choose the control law

$$v = -kz - B^T Px, \quad k > 0,$$

where P is the positive definite matrix from the Lyapunov equation above. Then the closed-loop system satisfies

$$\dot{V} \leq -\alpha\|x\|^2 + \beta|z|^2,$$

and

$$\dot{z} = -kz - B^T Px.$$

Consider the composite Lyapunov function

$$W(x, z) = V(x) + \frac{1}{2}z^2.$$

Its derivative is

$$\dot{W} = \dot{V} + z\dot{z} \leq -\alpha\|x\|^2 + \beta|z|^2 - kz^2 - zB^T Px.$$

Choosing k sufficiently large (e.g., $k > \beta + \frac{1}{2}\|B^T P\|^2$), we obtain

$$\dot{W} \leq -c_1\|x\|^2 - c_2z^2$$

for some positive constants c_1, c_2 . Hence the origin $(x, z) = (0, 0)$ is globally asymptotically stable.

Returning to the original variables,

$$v = -k(x_3 - x_3^*) - B^T P \begin{bmatrix} x_1 - r \\ x_2 \end{bmatrix}.$$

Finally,

$$u = \frac{3}{2}[(1 - x_1)x_3 + Tv].$$

The gain k can be selected to satisfy the input constraint $|u(t)| \leq 2$ for the specified initial condition.

Problem 8

Consider the TORA system (A.49)–(A.52) with

$$\Delta(x_1) = (J + mL^2)(m + M) - m^2L^2 \cos^2 x_1 > 0.$$

We apply the feedback

$$u = -\phi_1(x_1) - \phi_2(x_2),$$

where ϕ_i are locally Lipschitz, satisfy $\phi_i(0) = 0$, and

$$y\phi_i(y) > 0 \quad \forall y \neq 0.$$

Assume further that

$$\int_0^y \phi_1(s) ds \rightarrow \infty \quad \text{as } |y| \rightarrow \infty.$$

From Exercise 5.10, the system with input u and output x_2 is lossless with storage function

$$S(x) = \frac{1}{2}v^T D(x_1)v + \frac{1}{2}kx_3^2, \quad v = \begin{bmatrix} x_2 \\ x_4 \end{bmatrix},$$

satisfying

$$\dot{S} = ux_2.$$

Applying the preliminary feedback

$$u = -\phi_1(x_1) + w,$$

the augmented storage function

$$V(x) = S(x) + \int_0^{x_1} \phi_1(\lambda) d\lambda$$

satisfies

$$\dot{V} = wx_2.$$

Now choose

$$w = -\phi_2(x_2),$$

so that the final control law becomes

$$u = -\phi_1(x_1) - \phi_2(x_2),$$

and

$$\dot{V} = -x_2\phi_2(x_2) \leq 0.$$

Since $x_2\phi_2(x_2) > 0$ for $x_2 \neq 0$,

$$\dot{V} = 0 \iff x_2 = 0.$$

Let

$$E = \{x : \dot{V} = 0\} = \{x : x_2 = 0\}.$$

Suppose a trajectory remains in E . Then $x_2(t) \equiv 0$, and from (A.49),

$$\dot{x}_1 = 0 \implies x_1 = c.$$

From (A.50) with $x_2 = 0$ and $u = -\phi_1(c)$,

$$0 = \frac{1}{\Delta(c)} \left[(m + M)(-\phi_1(c)) + mLk \cos c x_3 \right].$$

From (A.52) with $x_2 = 0$ and $x_4 = 0$,

$$0 = \frac{1}{\Delta(c)} \left[mL\phi_1(c) \cos c - (J + mL^2)kx_3 \right].$$

Eliminating x_3 from these two equations yields

$$\phi_1(c)\Delta(c) = 0.$$

Since $\Delta(c) > 0$ for all c , we obtain

$$\phi_1(c) = 0.$$

Because $y\phi_1(y) > 0$ for $y \neq 0$, it follows that

$$c = 0.$$

Substituting back gives $x_3 = 0$, and from (A.51), $x_4 = 0$.

Hence the largest invariant set contained in E is

$$\{0\}.$$

The function $V(x)$ is positive definite. The term $S(x)$ is quadratic in (x_2, x_3, x_4) . The additional term

$$\int_0^{x_1} \phi_1(\lambda) d\lambda$$

is radially unbounded by assumption.

Hence V is radially unbounded.

By LaSalle's invariance principle, all trajectories converge to the largest invariant set in E , which is the origin.

Therefore, the origin is globally asymptotically stable under the feedback

$$u = -\phi_1(x_1) - \phi_2(x_2).$$